18-755: Networks in the Real World

Milestone 0: Project Proposal

Tentative project title: Social Connections and Consumer Behavior on Yelp

Project team member(s) and id(s): Lyson(Zixuan) Chen, zchen4; Xiaohua Zhang, xiaohuaz

Overview of the project topic (~5 sentences): This project studies how social network structure is related to consumer behavior on Yelp. We will represent Yelp users as a social network based on their friendship connections. We are interested in how social groups and user positions in the network relate to restaurant choices and review behavior. Understanding these patterns may help reveal meaningful consumer groups and important users in the network. The results may provide useful insights for customer segmentation, recommendation, and marketing.

Goal of the project (~5 sentences): We will study whether friendship communities represent different consumer groups with distinct restaurant, cuisine, or location preferences. We will also examine whether central users or bridge users show more diverse consumer behavior than other users. Our goal is to understand how network communities and user positions are related to consumer behavior. We hope to identify meaningful patterns that cannot be seen from individual reviews alone.

Dataset and networks (~6 sentences): We will use Yelp Open Dataset, the following is the URL: https://business.yelp.com/data/resources/open-dataset/. The dataset contains information about users, reviews, businesses, and user friendships. We will build an undirected social network where each node represents a Yelp user and each edge represents a friendship between two users. Review data will be used to describe each user's restaurant choices and rating behavior. Business data will provide information such as restaurant categories and locations. We will analyze community structure, degree, clustering, betweenness centrality, and other network properties to study how social structure is related to consumer behavior.

Division of labor (~5 sentences):
Lyson will build the friendship network and analyze how communities differ in consumer preferences. Xiaohua will build user behavior features and analyze whether central and bridge users show more diverse consumer behavior. Both members will contribute to the comparison between network structure and consumer behavior. We will work together on result interpretation, visualization, and writing. A shared GitHub repository and regular meetings will be used to track progress and keep the workload balanced.